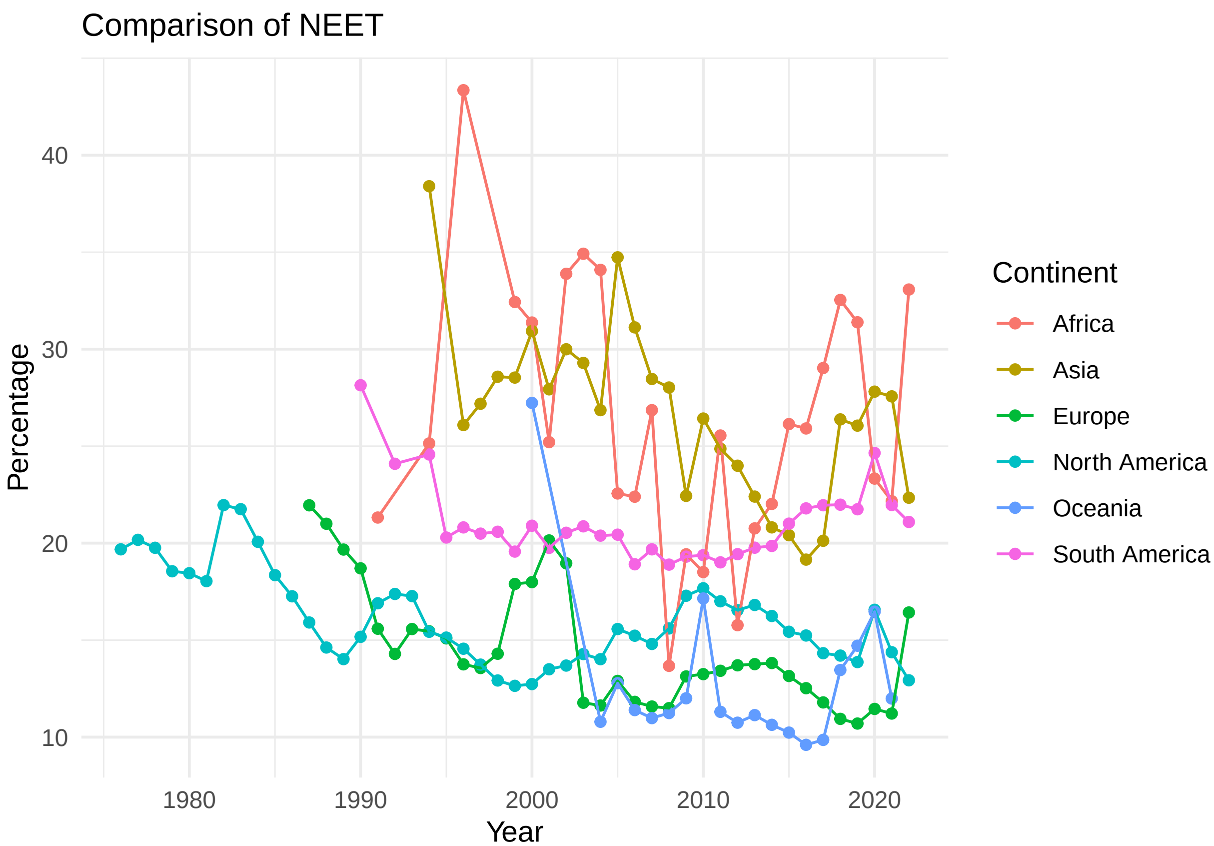
Introduction to Data Science Group Project Task 2

K9

2025-12-04

In this section, we will discuss the extent each continent working toward the goal of reducing the proportion of youth not in employment, education or training (hereafter referred to as NEET).

We would first draw a graph to demonstrate the change of NEET for each continent with the year using the line graph.

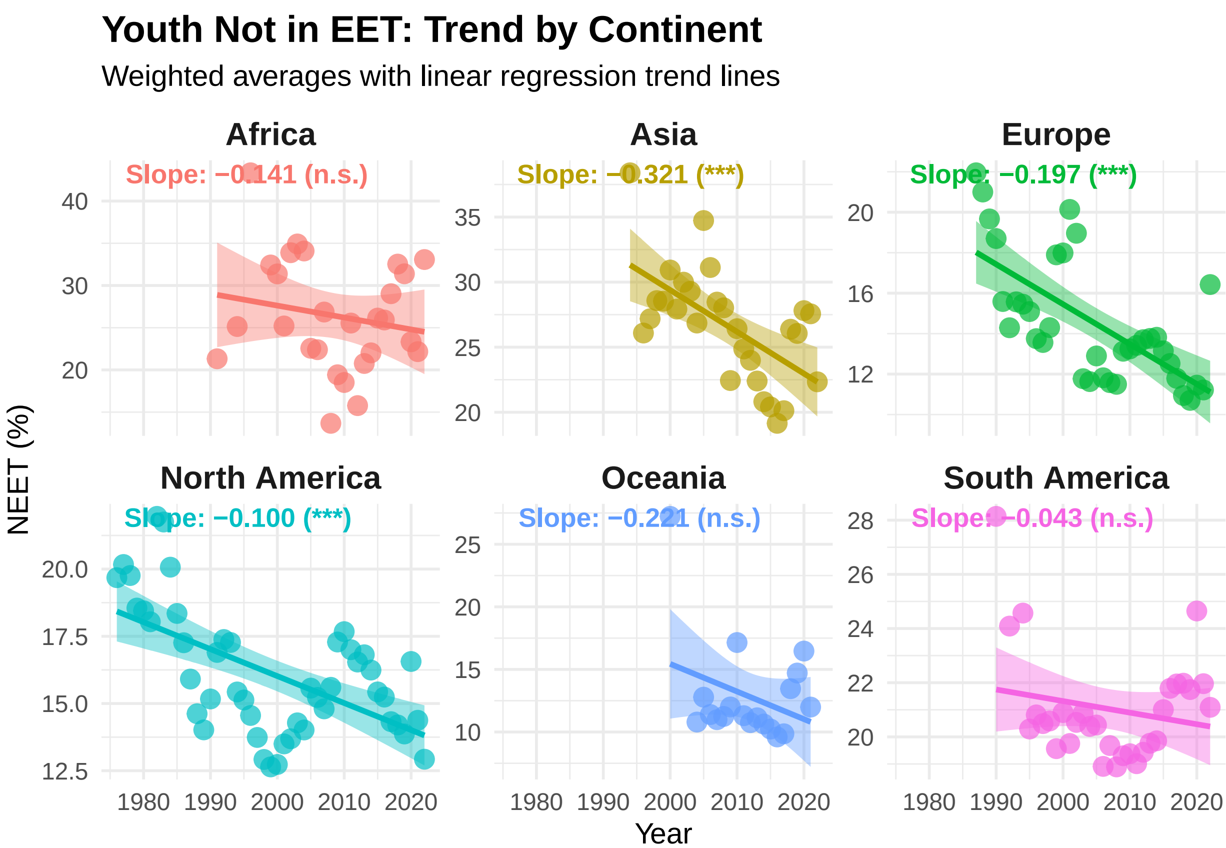


We then would like to see the regression line for the NEET.

Table 1: Hypothesis Testing Results: With and Without Outliers (IQR multiplier: 1.0)

Continent	Slope	P-value	Significance	Significant at 5%? (with)
Africa	-0.1413	0.3677	n.s.	No
Asia	-0.3211	0.0006	***	Yes
Europe	-0.1974	0.0000	***	Yes
North America	-0.1003	0.0000	***	Yes
Oceania	-0.2209	0.1783	n.s.	No
South America	-0.0427	0.2889	n.s.	No

```
## `geom_smooth()` using formula = 'y ~ x'
```



Then we would like to see how education and employment each contribute to the EET, one would create a table to illustrate how they correlated

Table 2: Correlation of NEET with Youth Unemployment and School Enrollment (Continental)

Continent	Corr_SE	P_Value_SE	Corr_Unemp	P_Value_Unemp
Africa	-0.0977	0.6411	0.6313	0.0005
Asia	-0.7247	0.0000	0.6190	0.0006
Europe	-0.5455	0.0080	0.2152	0.2359
North America	-0.2242	0.3144	0.7416	0.0000
Oceania	-0.7727	0.0081	-0.1909	0.4463
South America	0.3922	0.0590	0.3393	0.0672

If we do not group the data by continent rather seeing them as a whole,then we get the following table

Table 3: Correlation of NEET with Youth Unemployment and School Enrollment (World)

variable	correlation	p_value
SE	0.3161	0
Unemployment	-0.5574	0

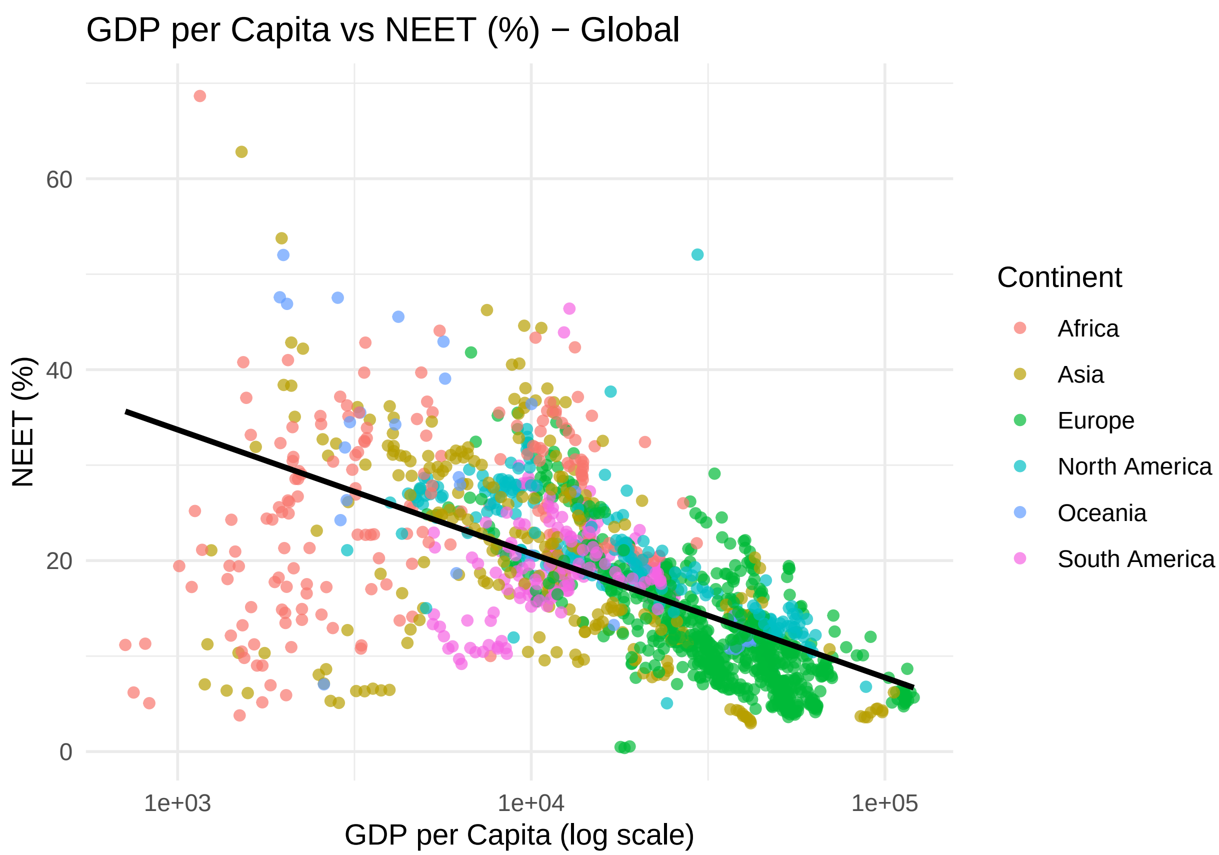
Global Correlation

```
##
##  Pearson's product-moment correlation
##
## data:  gdp_neet$Gdp_per_capita and gdp_neet$Percentage_NEET
## t = -32.692, df = 1550, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.6673823 -0.6084206
## sample estimates:
##        cor
## -0.6388386
```

```
##
##  Spearman's rank correlation rho
##
## data:  gdp_neet$Gdp_per_capita and gdp_neet$Percentage_NEET
## S = 1058310214, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.6985928
```

Global Plot

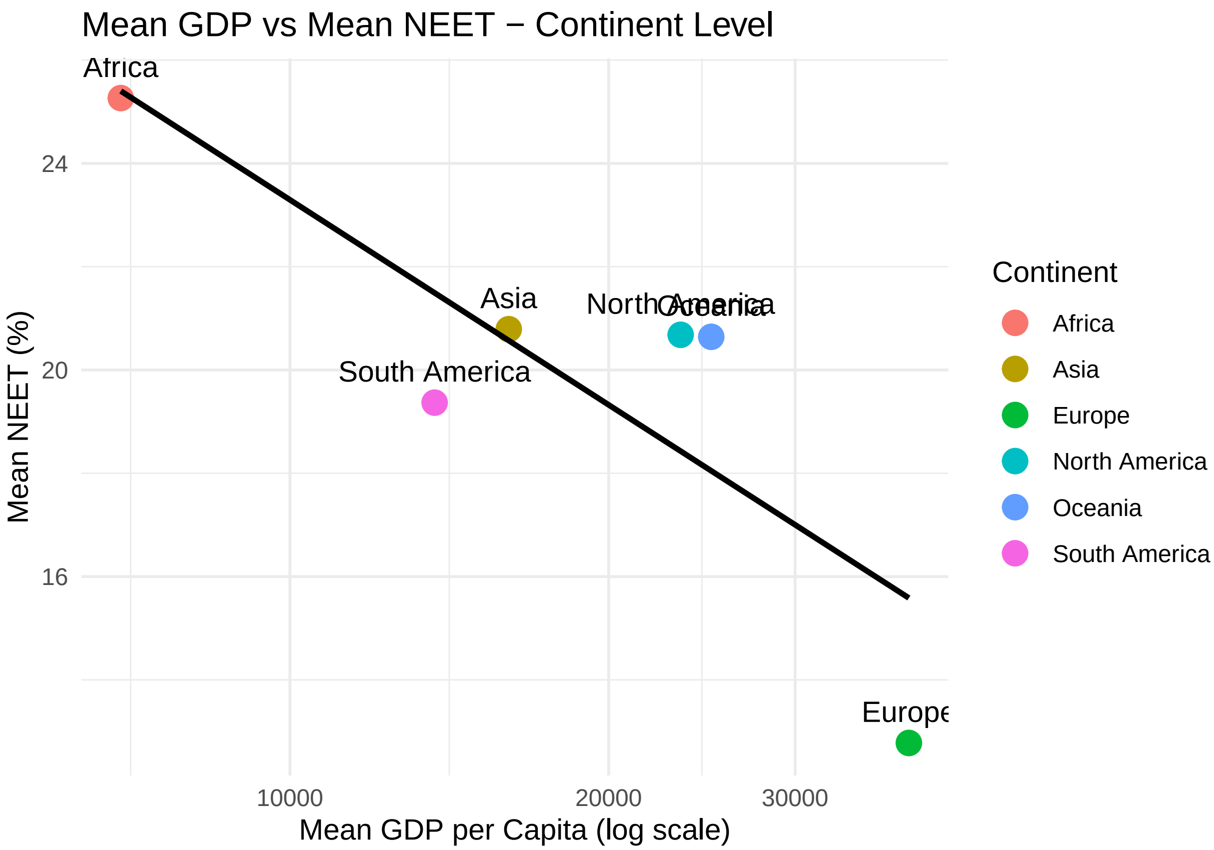
```
## `geom_smooth()` using formula = 'y ~ x'
```



Continent Means + Correlation

Continent Plot

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##                         Level  Pearson_r Spearman_r       p_value    N
## 1           Global (Countries) -0.6388386 -0.6985928 9.660679e-179 1552
## 2 Continent Means (6 points) -0.8816186 -0.6571429  2.019173e-02    6
##   Significant
## 1        TRUE
## 2        TRUE
```